

Family Name .....

Given Name .....

Student No. ....

Signature .....

**THE UNIVERSITY OF NEW SOUTH WALES**

**School of Electrical Engineering & Telecommunications**

**FINAL EXAMINATION**

**Term 1, 2020**

**ELEC1111**

**Electrical and Telecommunications Engineering**

**TIME ALLOWED: 2 hours**

**TOTAL MARKS: 100**

**TOTAL NUMBER OF QUESTIONS: 5**

**THIS EXAM CONTRIBUTES 50% TO THE TOTAL COURSE ASSESSMENT**

Reading Time: 10 minutes.

This paper contains 6 pages.

Candidates must **ATTEMPT ALL** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **MAY NOT** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

## QUESTION 1 [20 marks]

- a. **(10 marks)** For the circuit shown in Figure 1,
- (7 marks)** Apply mesh analysis and write down the mesh equations using the labels provided. Simplify the equations in the form  $ai_1 + bi_2 + ci_3 = d$ , where coefficients  $a, b, c$  and  $d$  can be positive or negative.  
Note: DO NOT solve the equations.
  - (3 marks)** Given the values of mesh currents as  $i_1 = 3.5\text{ A}$ ,  $i_2 = -0.5\text{ A}$  and  $i_3 = 2.5\text{ A}$ , find the power in the dependent voltage source.

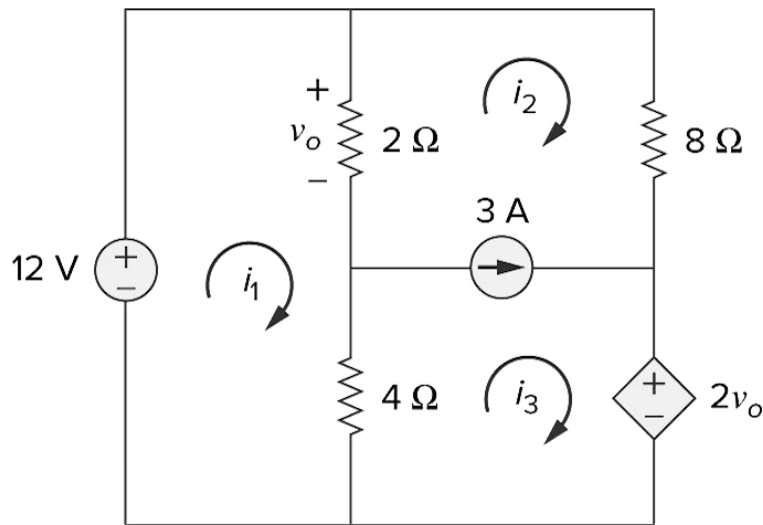


Figure 1

- b. **(10 marks)** Transistors can serve as amplifiers, producing both current gain and voltage gain. They can be used to furnish a considerable amount of power to devices that convert energy, such as loudspeakers or control motors. Calculate the gain (i.e.  $v_o/v_s$ ) of the transistor amplifier circuit shown in Figure 2 when  $v_s = 2\text{ V}$ .

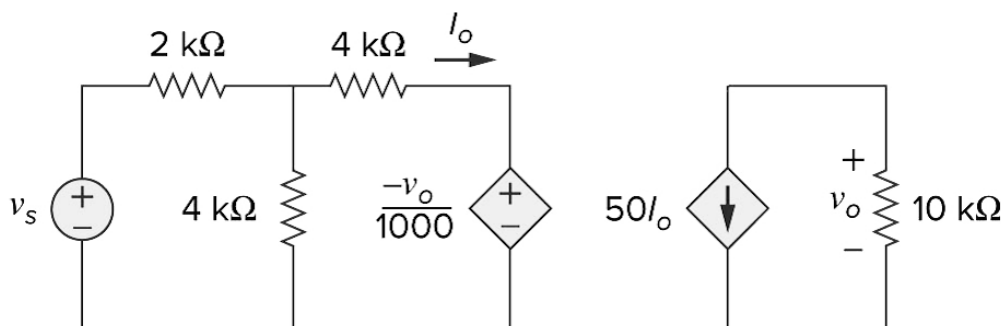


Figure 2

## QUESTION 2 [20 marks]

- a. **(8 marks)** A magnetically controlled switch is called a *relay*. A relay is essentially an electromagnetic device used to open or close a switch that controls another circuit (see Figure 3(a)). The coil circuit is an RL circuit like that in Figure 3(b), where  $R$  and  $L$  are the resistance and inductance of the coil.

A certain relay has a resistance of  $200\Omega$  and an inductance of  $500\text{ mH}$ . The relay contacts close when the current through the coil reaches  $175\text{ mA}$ . What time elapses between the application of  $110\text{V}$  to the coil (when  $S_1$  closes at  $t = 0$ ) and contact closure?

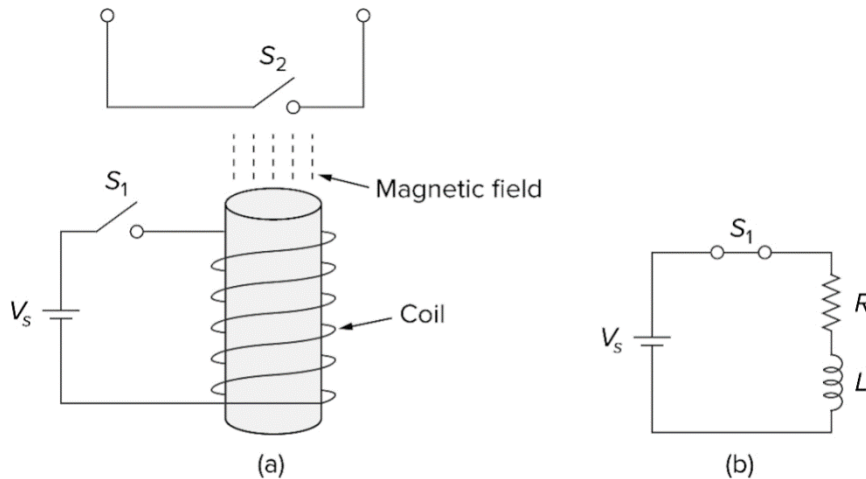


Figure 3

- b. **(12 marks)** For the circuit shown in Figure 4,
- (2.5 marks)** Find the initial voltage  $v(0^-)$  across the capacitor under steady-state condition.
  - (2.5 marks)** Find the final voltage  $v(\infty)$  across the capacitor under steady-state condition.
  - (4 marks)** Derive an expression for the voltage of the capacitor  $v(t)$  for all time (i.e., for both  $t < 0$  and  $t \geq 0$ ).
  - (3 marks)** Derive an expression for the current  $i(t)$  for  $t \geq 0$ .

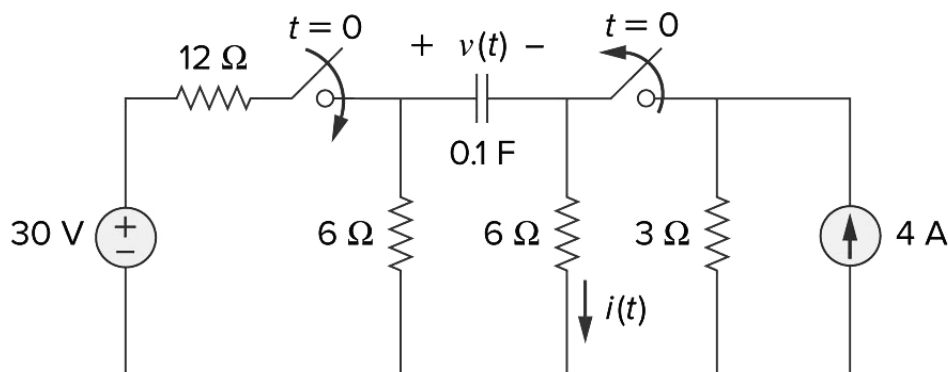


Figure 4

### QUESTION 3 [25 marks]

- a. (12 marks) In the circuit of Figure 5, calculate the voltage  $v_o$  if  $v_s = 2V$ .

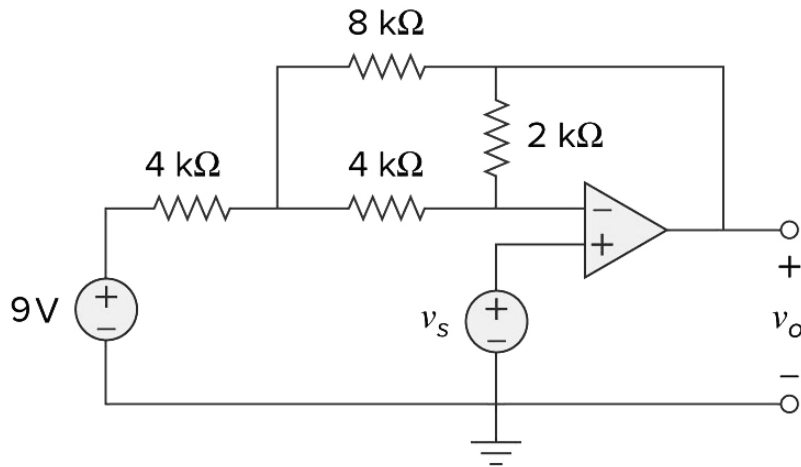


Figure 5

- b. (13 marks) Woofers or bass speakers are used to reproduce low frequency sounds, typically from 40 Hz up to 500 Hz. The circuit in Figure 6 is used to set up a woofer speaker using a Sallen-Key low pass filter. If  $V_{in}$  is an AC signal with RMS value  $V_{in_{RMS}} = 500 \text{ mV}$  and frequency  $f = 500 \text{ Hz}$ , calculate:
- (2 marks) Input voltage  $V_{in}$  as a function of time and then as a phasor.
  - (8 marks) Output voltage  $V_o$  of the Sallen-Key low pass filter as a phasor.
  - (3 marks) Average power absorbed by the woofer speaker (i.e.  $8\Omega$  resistor).

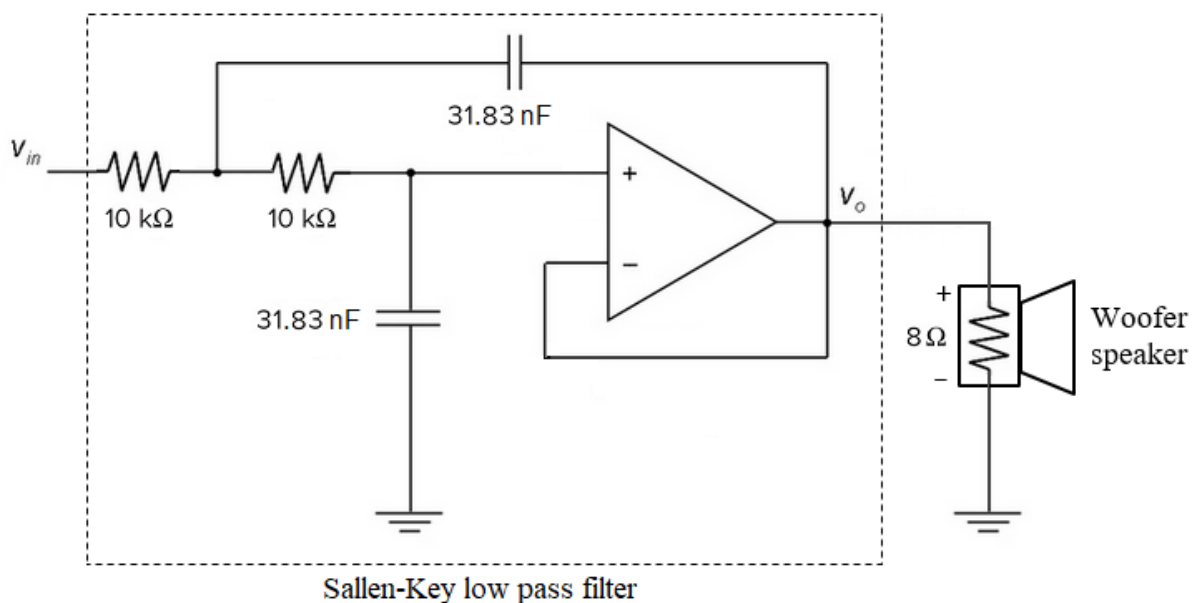


Figure 6

#### QUESTION 4 [25 marks]

- a. **(10 marks)** A phase shifting circuit is often employed to correct an undesirable phase shift already present in a circuit or to produce special desired effects. The circuit in Figure 7 shows an RC phase shift circuit with a certain lagging phase shift. Calculate the value of  $V_o$  as a function of  $V_i$  and state the phase shift introduced by the circuit.

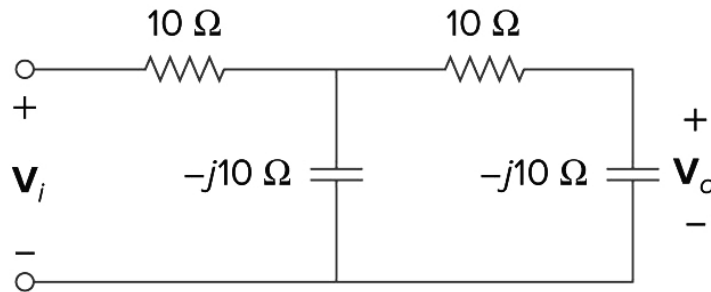


Figure 7

- b. **(15 marks)** For the circuit shown in Figure 8,
- (5 marks)** Calculate the load impedance  $Z_L$  that will ensure the maximum transfer of power from the circuit to the load.
  - (10 marks)** Find the maximum average power received by  $Z_L$ .

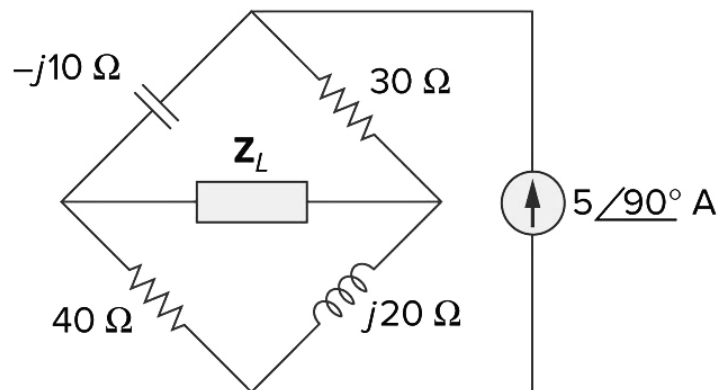


Figure 8

## IGNORE THIS QUESTION

### QUESTION 5 [10 marks]

- a. (4 marks) Simplify the following logical expression:

$$Z = \overline{A + \overline{B}C + \overline{C}}$$

- b. (6 marks) Consider the logical diagram shown in Figure 9.

- (1.5 marks) Derive the logical expression for  $Y$ .
- (4.5 marks) Construct the truth table relating  $Y$  to inputs  $A$ ,  $B$  and  $C$ .

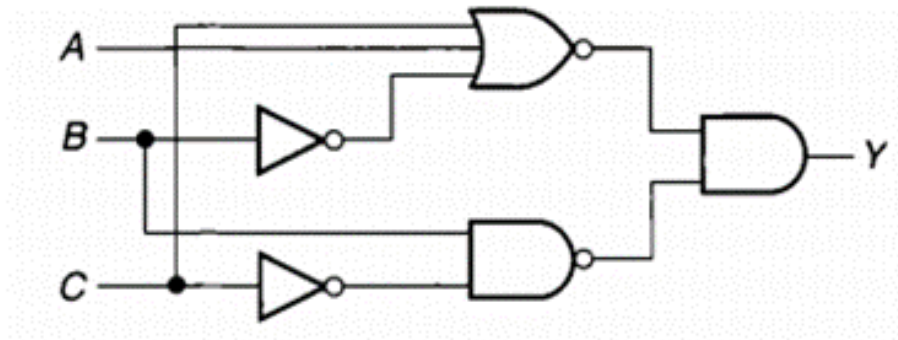


Figure 9